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A biologically inspired artificial neuron with intrinsic plasticity based on monolayer molybdenum disulfide

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Neuromorphic hardware that accurately simulates diverse neuronal behaviours could be of use in the development of edge intelligence. Hardware that incorporates synaptic plasticity—adaptive changes that strengthen or weaken synaptic connections—has been explored, but mimicking the full spectrum of learning and memory processes requires the interplay of multiple plasticity mechanisms including intrinsic plasticity. Here we show that an integrate-and-fire neuron can be created by combining a dynamic random-access memory and an inverter that are based on wafer-scale monolayer molybdenum disulfide films. In the system, the voltage in the dynamic random-access memory capacitor—that is, the neuronal membrane potential—can be modulated to emulate intrinsic plasticity. The module can also emulate the photopic and scotopic adaptation of the human visual system by dynamically adjusting its light sensitivity. We fabricate a 3 × 3 photoreceptor neuron array and demonstrate light coding and visual adaptation. We also use the neuron module to simulate a bioinspired neural network model for image recognition.

With increases in the complexity of measurement scenarios and amounts of data, the development of edge intelligence hardware faces challenges in terms of speed and power consumption. Biological neural systems exhibit higher efficiency and lower power consumption in perceiving and recognizing complex environments than existing artificial vision systems[1,2]. However, retinomorphic sensors, together with biologically inspired neural networks (BioNNs[3]), could be a solution to increase efficiency and reduce the power consumption of artificial vision systems.

The transfer of information between neurons using electrical spikes is the most approximate form of biological neural activity, and the precise temporal processing of electrical spikes is a critical strategy for encoding information in BioNNs[4]. However, current spike coding schemes based on complementary metal–oxide–semiconductor circuits consume considerable hardware resources[5], and the incorporation of temporal sparsity coding strategies into neuromorphic hardware remains challenging. The human visual system has the ability to preprocess the visual stimuli it receives before sending them to the temporal lobe for high-level processing such as learning and recognition. When applied to an artificial visual system, the visual adaptation function of human retinal cells could reduce hardware complexity and power consumption.

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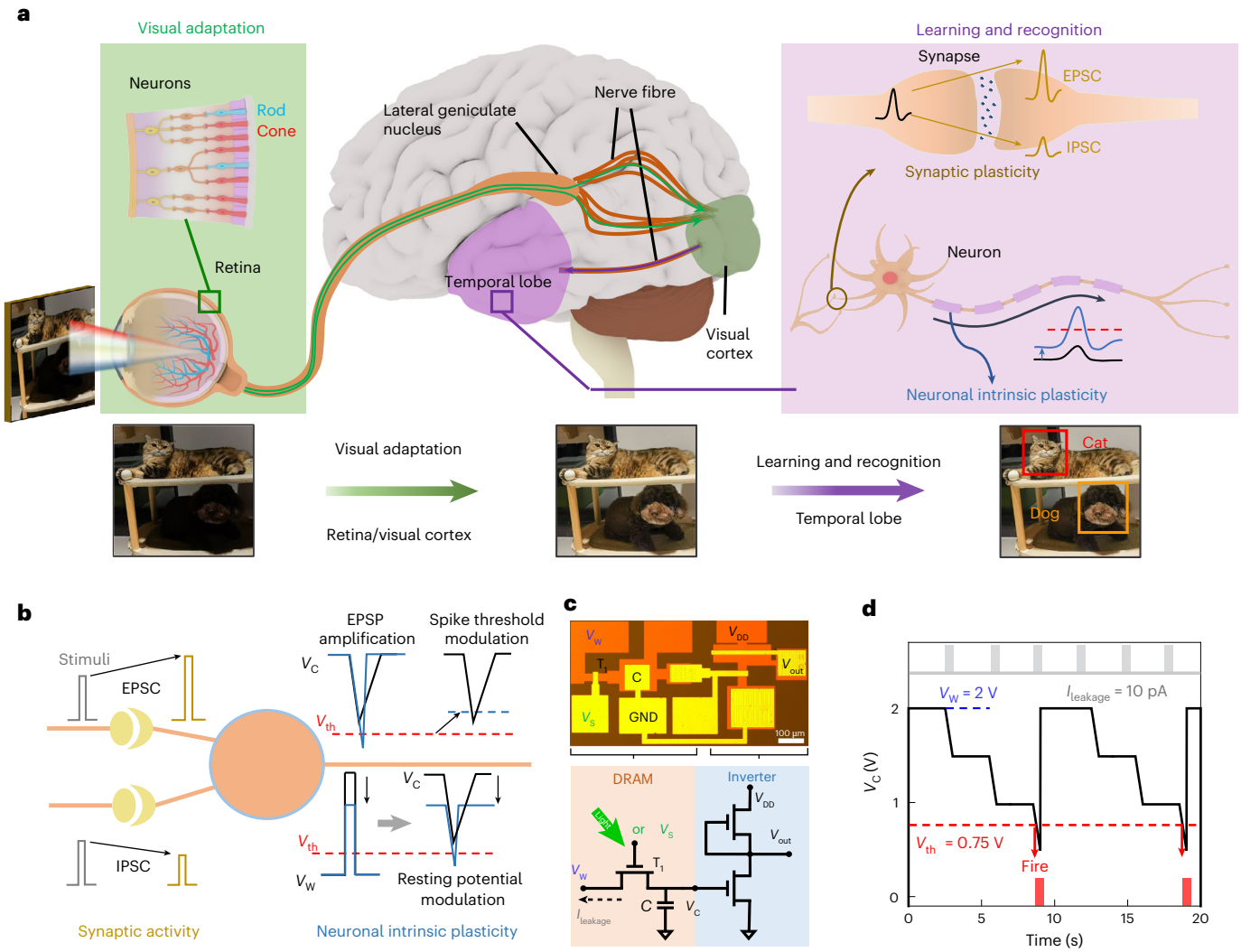


Fig. 1 | Versatile neuron module design. a, Process of visual adaptation in the human retina and feature recognition in the cerebral cortex is a complex phenomenon. Photoreceptor cells in the human retina encode light stimuli as electrical spike signals. These signals are relayed by the lateral geniculate nucleus and transmitted to the visual cortex. Visual adaptation occurs in the retina by adjusting the activity of cone and rod cells. The temporal lobe is responsible for learning and recognizing features through synaptic and neuronal plasticity. EPSC, excitatory postsynaptic current; IPSC, inhibitory postsynaptic current. Right inset: the black and blue curves denote spikes before and after intrinsic plasticity adjustment, respectively. The red dashed line is the firing threshold. **b**, Mechanisms of synaptic activity and neuronal intrinsic plasticity. Neurons can facilitate firing by increasing the EPSP amplitude, adjusting the firing threshold V_{th}, or adjusting the initial position of the membrane potential V_C (that is, resting potential). In our neuron circuit, resting potential modulation is achieved via the input signal V_W. **c**, Optical microscopic image and circuit diagram of the versatile neuron device consisting of a DRAM and an inverter based on a MoS_2 transistor. T_1, the access transistor in DRAM; V_W: the input signal for controlling the resting potential; V_S: the gate control signal of T_1. The dashed black arrow represents the leakage of T_1, which is affected by both V_S or light intensity. **d**, IF process performed by the DRAM inverter neuron module. The top grey pulses and bottom red pulses indicate the input timing of V_S to the neuron and the neuron's firing timing, respectively. The initial membrane potential, set by V_W, gradually decreases due to T_1 transistor leakage from each V_S input, until reaching the firing threshold V_{th} and triggering an output.

High-level visual processing is associated with neural activities, including the plasticity regulation of synaptic and neuronal intrinsic features, which are essential for learning, memory and recognition. Diverse mechanisms of synaptic plasticity—including long-term potentiation, short-term plasticity and spike-timing-dependent plasticity (STDP)—have been investigated using a variety of synaptic devices such as resistive random-access memory, phase-change memory and ion transistors[6–10]. However, synaptic plasticity alone does not elucidate the full spectrum of learning and memory processes. Several phenomena associated with changes in neuronal membrane potential—including the amplification of excitatory postsynaptic potentials (EPSPs), adjustments in spike threshold and changes in resting potential—play a role in the transmission and processing of neuronal signals[11,12]. These intrinsic plasticity phenomena are present in almost all types of neuron[12–16], and are integral to the learning and adaptability of the biological neural system.

There have been some initial explorations of hardware implementations of neural intrinsic plasticity. A hybrid complementary metal–oxide–semiconductor/resistive random-access memory neuron circuit that can adaptively regulate the input stimuli for the precise control of spike firing rates by emulating the intrinsic plasticity of neurons has been developed[15,17]. A neurotransistor that simulates the nonlinear integration process of neurons and achieves intrinsic plasticity by adjusting the spike firing threshold has also been created[10,16,17]. Intrinsic plasticity-based neuron oscillatory behaviour and spike threshold modulation using phase-change memory devices has been reported[10,17]. These advances highlight the potential of intrinsic plasticity in building BioNN computing architectures, particularly in terms of enhancing the

learning stability and environmental adaptability of neural networks. However, to effectively and reliably exploit the intrinsic plasticity of biological neurons, the development of simplified neuromorphic hardware with improved stability, efficiency and wafer-level integration is required.

In this Article, we report a neuron module that combines a dynamic random-access memory (DRAM) and an inverter that are based on wafer-scale monolayer molybdenum disulfide (MoS_2) films. Two-dimensional semiconductors, such as MoS_2, have efficient light–matter interactions, wide spectral range of optical absorption and highly tunable optoelectronic properties. In recent years, progress has been made in the wafer-scale growth of two-dimensional semiconductor materials and the fabrication of small-scale integrated circuits[18–26], driving the development of edge intelligence based on two-dimensional semiconductors towards the system level. By exploiting the tunable optoelectronic properties of MoS_2, our neuron module exhibits photopic and scotopic adaptation similar to the human visual system. We also use the approach to create a BioNN circuit that can emulate the intrinsic plasticity of the neuronal membrane resting potential and perform time-to-first spike (TTFS) encoding. Finally, we simulate a BioNN model for image recognition using this neuron module and show that it could simplify or reduce circuitry and processing algorithm requirements for implementing biologically inspired visual systems in hardware.

Versatile neuron module design

Figure 1a shows the process of visual adaptation in the human retina and feature recognition in the cerebral cortex. The photoreceptor cells in the human retina convert light stimuli into electrical signals and are categorized into two main types: cone cells and rod cells. Rod cells play a crucial role in scotopic vision, enabling humans to see in dim environments. Cone cells are responsible for photopic vision, which allows humans to see in bright light and distinguish between different colours[27–29]. The activity of cones and rods is regulated by the horizontal cells to adjust light sensitivity. The encoded electrical spike signals are transmitted to the visual cortex after relaying through the lateral geniculate nucleus. The visual cortex transmits the received visual information through nerve fibres to the temporal lobe for high-level processing, including the recognition of biotic or abiotic features. Recent studies have shown that non-synaptic plasticity, such as neuronal intrinsic plasticity, plays a crucial role in coordinating complex brain functions[11,30], in addition to the widely regarded synaptic plasticity mechanism responsible for learning and memory. The input–output functions modulated by synaptic activity and intrinsic plasticity are shown in Fig. 1b. Synaptic activity merely reflects the degree of signal amplification or inhibition of signals. Intrinsic plasticity is achieved through three mechanisms: amplification of EPSP, adjustment of spike threshold and alteration of resting membrane potential. To simulate the last kind of intrinsic plasticity, we need to construct circuit elements with the fundamental properties of integrate-and-fire (IF) neurons.

A versatile artificial neuron module was designed and fabricated for optical encoding, visual adaptation and simulating neuronal intrinsic plasticity. Figure 1c illustrates the optical microscopic image and circuit schematic of the neuron module. This module consists of a DRAM and an inverter based on a MoS_2 field-effect transistor. The DRAM cell functions as an integrator, with the MoS_2 transistor (T_1) regulating the writing or erasing of charges on the capacitor. The inverter's input is connected to a capacitor, which provides the voltage necessary to achieve the threshold firing function. The capacitance C of the DRAM was determined to be approximately 29 pF (Supplementary Fig. 3). The basic characteristics of the MoS_2 transistor and inverter are provided in Supplementary Note 1 and Supplementary Figs. 4–6. Additionally, we constructed a 3 × 3 BioNN circuit utilizing the versatile neuron module (Supplementary Fig. 1). Furthermore, Supplementary Fig. 2 illustrates an extended BioNN that incorporates versatile neurons with different functions.

Our versatile neuron module should function as a neuron with IF dynamics. The potential across the capacitor (C) represents the membrane potential (V_C) of the neuron cell. To reset the neuron module, a positive selected voltage (V_S) is applied to the T1 gate to turn on the transistor. Meanwhile, a write voltage (V_W) is written to the capacitor through the drain of T_1, resetting the membrane potential to its resting potential. The charges written to the capacitor, Q_{reset}, can be calculated using $V_W \times C$.

The neuron's output can be maintained at a low voltage level by applying a negative V_S when there are no input stimuli. At this point, appropriate optical or electrical signals were applied to drive the transistor T_1 into the subthreshold region, thereby regulating the leakage across T_1 to an appropriate level. As a result, V_C gradually decreased as the stored charges were discharged through T_1. During the above process, the number of charges on the capacitor over time ($Q(t)$) can be mathematically described by the following equation:

$$Q(t) = Q_{reset} - \int_0^t I_{leakage}(t)\mathrm{d}t. \quad (1)$$

Since $Q(t)$ is equal to $V_C \times C$, the membrane potential of the IF neuron in the DRAM inverter can be expressed as

$$V_C = V_W - \frac{1}{C}\int_0^t I_{leakage}(t)\mathrm{d}t. \quad (2)$$

The state equation corresponds to the canonical equation of the IF neuron but has a modification in the direction of the integration process. Specifically, in the conventional IF neuron model[31], the charging current flowing into the capacitor is integrated, causing the membrane potential to gradually rise towards the firing threshold. By contrast, our IF neuron design integrates the discharging current in the opposite direction. Equation (2) was used to accurately reproduce the integration process described above. Figure 1d shows a complete simulation of the IF process using parameters extracted from our experimental measurements. The average $I_{leakage}$ and capacitance C were set at 10 pA and 30 pF, respectively, whereas V_W and time delay of reset (t_{delay}) were set to 2 V and 500 ms, respectively. A series of optical or electrical stimulations of T_1 produces $I_{leakage}$, which causes V_C to rapidly drop to the threshold voltage of the inverter (V_{th}), resulting in the switching of the inverter. After a delay, V_C was then recharged to the resting potential of V_W = 2 V, producing an output spike with a width equal to t_{delay}. The circuit diagram and time sequence of the peripheral reset circuit are shown in Supplementary Fig. 7, which consists of a reset transistor, a D-type flip-flop and a two-stage inverter. As demonstrated in Supplementary Fig. 8, an additional 42.7 μW of power is consumed via this circuit. Typically, the neuron will undergo a refractory period, and the corresponding output spike waveforms of the neuronal encoding process are displayed in Supplementary Fig. 9.

Synaptic activity and neuronal intrinsic plasticity

Figure 2a shows the input and output waveforms during the IF process of the fabricated neuron module. The reset process was initiated by applying a 3-V pulse to V_S and a 2-V pulse to V_W, which refreshed the membrane potential V_C to its resting level. Subsequently, during the integration process, a pulse sequence of −0.6 V with a pulse width of 500 ms was applied to V_S, resulting in a stepwise decrease in V_C until the neuron fired at the timing of the first output spike (t_{fire}). The experimental result is basically consistent with the simulation result in Fig. 1d. As the voltage of the V_S pulse increased, the IF neurons demonstrated the ability to encode 5-ms pulses, approaching the action potential pulse width of biological neurons (Supplementary Fig. 10).

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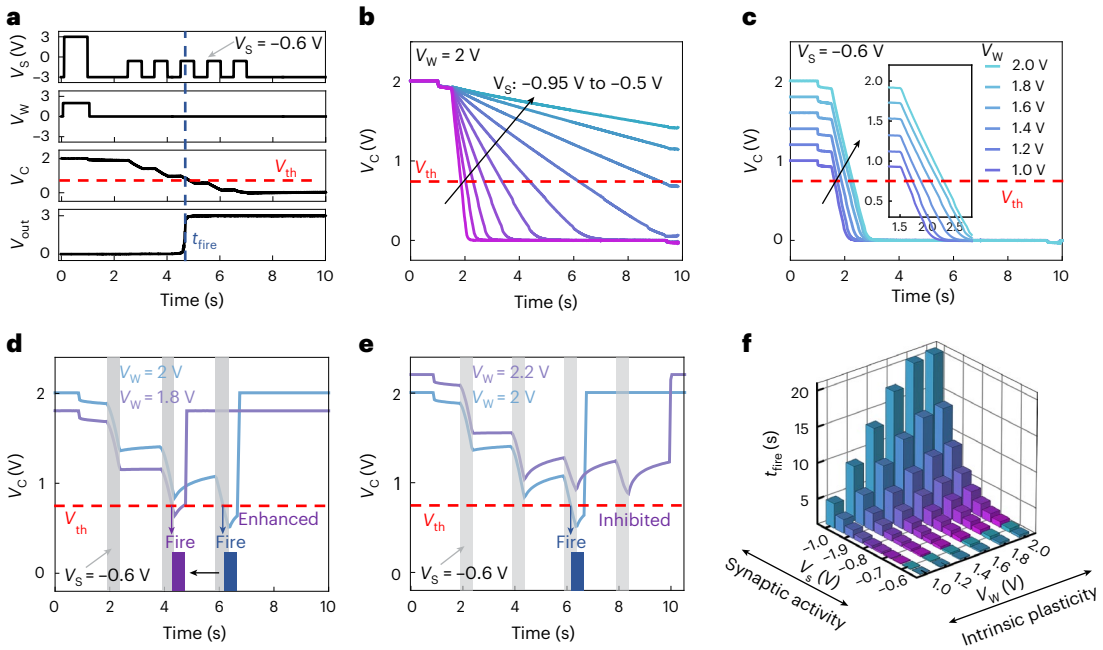


Fig. 2 | Synaptic activity and neuronal intrinsic plasticity of the neuron device. **a**, Electrical pulse sequence input and output waveforms during the IF process. **b**, Time-dependent V_C under various V_S from −0.95 V to −0.5 V with a step of 0.05 V, whereas V_W is fixed at 2 V. The black arrows indicate the V_C curve's trend when V_W shifts from −0.95 V to −0.5 V. **c**, Time-dependent V_C under various V_W from 1 V to 2 V with a step of 0.2 V, and V_S is fixed at −0.6 V. The black arrows also show the V_C curve's trend when V_S increases from 1.0 V to 2.0 V. **d**,**e**, Demonstrations of enhancement (**d**) and inhibition (**e**) of the neuronal intrinsic plasticity based on our neuron module. The purple and blue fire bars represent the neuron's firing timing with different V_W. The enhanced neuron fires earlier, whereas the inhibited neuron fires later or may not fire within the limited time frame. The grey-shaded regions indicate the input timing of V_S to the neuron. **f**, Modulation of t_{fire} obtained on the same device realized by adjusting the synaptic activity and intrinsic plasticity, represented by V_S and V_W, respectively.

Now that we have established the basic properties of IF neurons, our next step is to investigate the intrinsic plasticity of this neuron module. The advance or retardation of t_{fire} can be modulated by both V_S and V_W. Specifically, different EPSPs could be generated by modulating V_S, which is similar to the effect caused by different synaptic activities. On the other hand, changes in the neuronal resting potential are a critical expression of non-synaptic plasticity. Modulation is global and can potentially affect all synaptic inputs[11]. In our neuron module, the V_W written into the capacitor served as the resting potential and achieved the corresponding function.

Figure 2b depicts the time-dependent V_C under varying V_S (from −0.95 V to −0.5 V with a step of 0.05 V). As the intensity of V_S increases, V_C drops with a steeper slope, and t_{fire} increases. V_S actually modulates the amount of leakage in the DRAM transistor, which is analogous to modulating the intensity of the stimulus input to the neuron, that is, the effect produced by changes in synaptic activity. Similar effects can be achieved by modulating the intrinsic plasticity modulation. Figure 2c shows the time-dependent V_C under varying V_W (ranging from 1 V to 2 V with a step of 0.2 V), whereas V_S is fixed at −0.6 V. The starting point of the integration process is determined by V_W, which, in turn, affects the final t_{fire}.

Figure 2d,e elaborates the enhancement and inhibition of neuronal intrinsic plasticity. The neuron fires earlier with a lower V_W because the integration process starts at a lower resting membrane potential level. Conversely, the probability of the neuron firing within a given period (for example, 10 s) decreases with increasing V_W. As shown in Fig. 2f, varying synaptic activities and resting membrane potentials, simulated by changing V_S and V_W, respectively, can result in a wide range of t_{fire} modulation values. The corresponding specific test results are shown in Supplementary Figs. 11 and 12.

Additionally, the time-dependent V_C was measured under varying V_W (ranging from 1 V to 2 V with a step of 0.2 V) and V_S (ranging from −0.95 V to −0.5 V with a step of 0.05 V). The results are displayed in Supplementary Fig. 13, which illustrates the effect of the resting potential V_W and input pulse voltage (V_S) on the integration process. Supplementary Fig. 14 depicts the extracted time-dependent V_C under V_S = −0.5 V, demonstrating t_{fire} with millisecond-level temporal resolution. We also investigated the output of our neuron device under complex input patterns (Supplementary Figs. 15 and 16). These experimental results demonstrate that our neuron module exhibits spatiotemporal correlativity between its input and output[32,33]. The energy consumption of the neuron cell per spike is primarily attributable to three factors: the energy to refresh the DRAM, standby energy of T_1 during integration and the static energy consumption of the inverter during both integration and firing operations. For a TTFS encoding period of 100 ms, the energy consumption per spike is 2.82 nJ, which is comparatively low in relation to other technologies implementing TTFS coding at the same pitch (Supplementary Table 1).

Each neuron has multiple dendrites to receive input signals from various presynaptic neurons, whereas a single axon is responsible for sending output signals[34]. These output signals can be selectively transmitted to specific postsynaptic neurons connected to them. To achieve this parallel multi-input and multi-output pattern of neurons, we proposed a circuit connection mode (Supplementary Fig. 17) in which multiple transistors were added.

TTFS coding of the light signal

In our design, the timing of the first output spike contains all the necessary information for the subsequent training and learning of the BioNN. This coding strategy, known as TTFS coding, is more efficient than the widely used firing-rate-based coding[35]. We have realized the TTFS coding of the light signal with temporal sparsity by exploiting the modulation of intrinsic plasticity and the photosensitive properties of MoS_2.

Figure 3a shows the three-dimensional schematic and circuit of our neuron device, which acts as a photoreceptor. It receives input signals (V_S, V_W and a light signal with a wavelength of 550 nm) and

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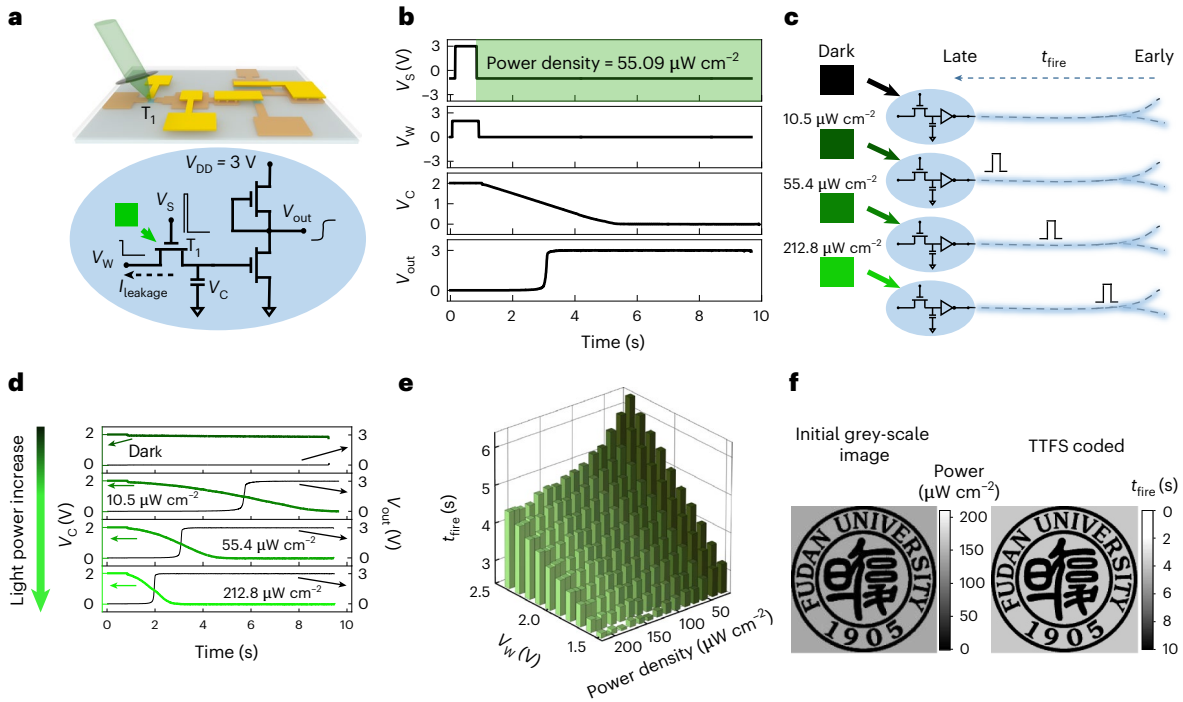


Fig. 3 | Light sensing and TTFS coding. a, Three-dimensional schematic and circuit diagram of our neuron module that acts as a photoreceptor, along with the waveforms of input signals with light stimuli (green block and arrow). **b**, Time sequence diagram of the input and output signals during the measurement. The incident light power density is 55.09 μW cm^{-2}. **c**, Demonstration of TTFS coding based on our photoreceptor neuron device. **d**, Schematic of the experimental measurement for TTFS coding with different light stimuli. **e**, Firing times of a neuron at various light intensities and writing voltage V_W (extracted from the data shown in Supplementary Fig. 20). **f**, Grey-scale image of the initial incident light intensity (left) and the image coded with t_{fire} using our time-to-first coding strategy (right).

converts light stimuli into electrical signals. Figure 3b displays the input and output waveforms. V_S was set to +3 V (reset state) after V_W rose to +2 V and was then fixed at −1 V (operating state) when V_W dropped to 0 V. The incident light intensity was set to 55.09 μW cm^{-2}. The photocurrent generated in T_1 under illumination contributes to a light-induced $I_{leakage}$. As shown in Supplementary Fig. 18, the membrane potential V_C gradually and linearly decreased due to the stable $I_{leakage}$ of T_1 in the saturation region (Supplementary Fig. 5) until the switching of the inverter was successfully triggered and emitted a spike when V_C dropped to 0.75 V.

The operation of the TTFS program on light stimuli is shown in Fig. 3c. To quantitatively evaluate the potential of our spike coding strategy, a shutter was used to segment the illumination into 10-s periods, which was considered as the temporal resolution. Our photoreceptors sampled the light intensity signal at each temporal resolution, determining the accuracy of the detection of dynamic events. After firing a spike, neurons remain in a refractory period until the next reset operation. Therefore, neurons could only fire one spike in a given temporal resolution. Brighter conditions lead to earlier firing due to increased $I_{leakage}$, whereas dimmer conditions result in delayed or no firing, as confirmed by the experimental results shown in Fig. 3d. Furthermore, to investigate the coding performance of the photoreceptor neuron, we conducted experiments to measure the membrane potential V_C under different conditions. These conditions included varying the incident light intensity and resting voltage V_W (ranging from +1.5 V to +2.4 V with a step of +0.1 V; Supplementary Fig. 19). Supplementary Fig. 20 shows the integration process of the neuron under the combined influence of both light power density and resting potential V_W. The resulting data were analysed to extract simulated t_{fire}, which were correlated with the incident light intensity and write voltage V_W (Fig. 3e).

Finally, the original grey-scale image of the incident light intensity (left) and the image coded with t_{fire} using TTFS coding (right) are displayed in Fig. 3f, demonstrating that our neuron device can successfully reproduce the original image.

Photopic and scotopic adaptation

The photopic and scotopic adaptation functions of cone cells and rod cells can be emulated by modulating light sensitivity through the V_S of our device (Fig. 4a). Specifically, the photopic adaptation of cone cells under bright conditions was simulated by reducing V_S to a low level, whereas the scotopic adaptation of rod cells under dim conditions was simulated by increasing the $I_{leakage}$ through a higher V_S. The photopic and scotopic adaptation processes of our photoreceptor neurons are illustrated in Fig. 4b. Both overbright and overdark images can be effectively identified with an appropriate V_S level. Supplementary Fig. 21 illustrates the 40 different devices prepared for light encoding and sensitivity tuning by adjusting V_S at a light intensity of 196.6 μW cm^{-2}. The t_{fire} distributions at both light sensitivities follow a focused Gaussian distribution, indicating that the devices exhibit excellent device-to-device uniformity.

Figure 4c illustrates the working mechanism during photopic adaptation under bright conditions. The original incident light intensities were mapped to four pixels, which were easily identifiable. When V_S = −1 V was applied, the high sensitivity of the photoreceptor neuron resulted in a sharp drop in V_C and ultimately an early t_{fire} with a poor degree of resolution. As a result, all four pixels were too bright to be identified. By decreasing the voltage to −1.4 V, the pixels were clearly distinguishable due to the photoreceptor neuron switching to lower sensitivity and V_C dropping more slowly. Figure 4d explains the working mechanism of our photoreceptor neuron during scotopic adaptation under dim conditions. This is in contrast to the situation under bright

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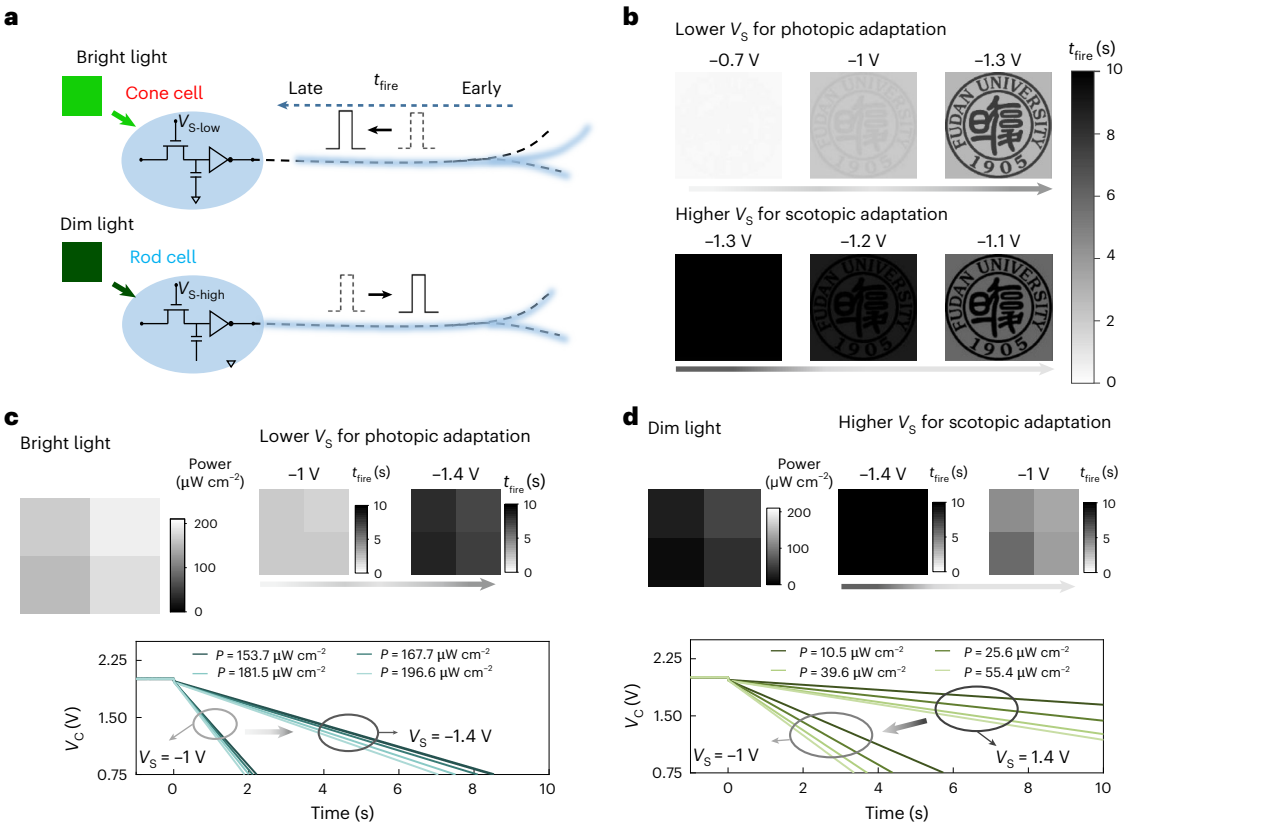


Fig. 4 | Photopic and scotopic adaptation by adjusting V_S. a, Schematic of photopic and scotopic adaptation. The light sensitivity of our photoreceptor neuron can be adjusted by modifying V_S, resulting in a tunable t_{fire}. **b**, Demonstration of the photopic and scotopic adaptation processes performed by our photoreceptor neuron under different conditions. Both overbright and overdark images can be effectively identified with an appropriate V_S level. **c**,**d**, Working mechanism of our device during photopic adaptation under bright conditions (**c**) and scotopic adaptation under dim conditions (**d**).

conditions. At a V_S of −1.4 V, the neuron was unable to fire with any temporal resolution, making it impossible to distinguish the mapping of t_{fire}. By changing V_S to −1 V, discrimination was ideally achieved due to reduced light sensitivity. Our photoreceptor neuron successfully mimicked the functions of cone and rod cells during both photopic and scotopic adaptation processes.

BioNN circuit for visual adaptation and feature recognition

We then construct a complete BioNN architecture for in situ image preprocessing and feature recognition. This architecture combines the fabricated 3 × 3 photoreceptor arrays as the preprocessing layer with a simulated neuron network based on neurons with intrinsic plasticity as the computing layer (Fig. 5a).

The 3 × 3 photoreceptor arrays are capable of achieving scotopic and photopic visual adaptation to both dim and bright light through the global tuning of V_S (Supplementary Figs. 22 and 23). The temporal information encoded in the preprocessing layer was subsequently input to the computational layer for the purpose of learning and feature recognition. The BioNN uses a hybrid control strategy, with the preprocessing layer using global control to adjust the light sensitivity, and the computing layer locally tuning the resting potentials of each single neuron (Supplementary Fig. 24). Given the temporal coding used, it is necessary to convert the timing information into digital data during the read-out process. Consequently, a read-out scheme that directly uses a time-to-digital converter was selected, as opposed to the conventional sample-and-hold with analogue-to-digital converter read-out circuit. The underlying principle of the counter-based time-to-digital converter is illustrated in Supplementary Fig. 25. The counter and control logic circuits that constitute the time-to-digital converter are implemented using a field-programmable gate array, with specific hardware resource utilization depicted in Supplementary Fig. 26.

Unsupervised learning via STDP using neuronal intrinsic plasticity

To fully leverage the potential of neuronal intrinsic plasticity in learning and inference tasks, it is crucial to develop effective and interpretable learning algorithms. Neuronal intrinsic plasticity requires developments in both hardware and algorithms. In this study, we adopted the learning rule from synaptic plasticity. Recent studies suggest that the STDP learning rule for synaptic plasticity may also be applied to neuronal intrinsic plasticity, too[11].

The STDP learning rule allows for the quantitative modification of synaptic connections based on the spike sequence of pre- and post-synaptic neurons[35–38]. The STDP learning rule for our neuron device was obtained through a simple mathematical derivation. Given the nearly linear relationship between V_W and t_{fire}, a linear normalization conversion relationship was established between the weight (W) and V_W:

$$W = 1 - \frac{V_W - \min(V_W)}{\max(V_W) - \min(V_W)}. \quad (3)$$

The $\max(V_W)$ and $\min(V_W)$ represent the maximum and minimum values of the resting potential V_W, respectively. In our simulation, these values were 3 V and 1 V, respectively. We substituted the weight update $\Delta W = W_2 - W_1$ into the classical STDP learning rules, and then

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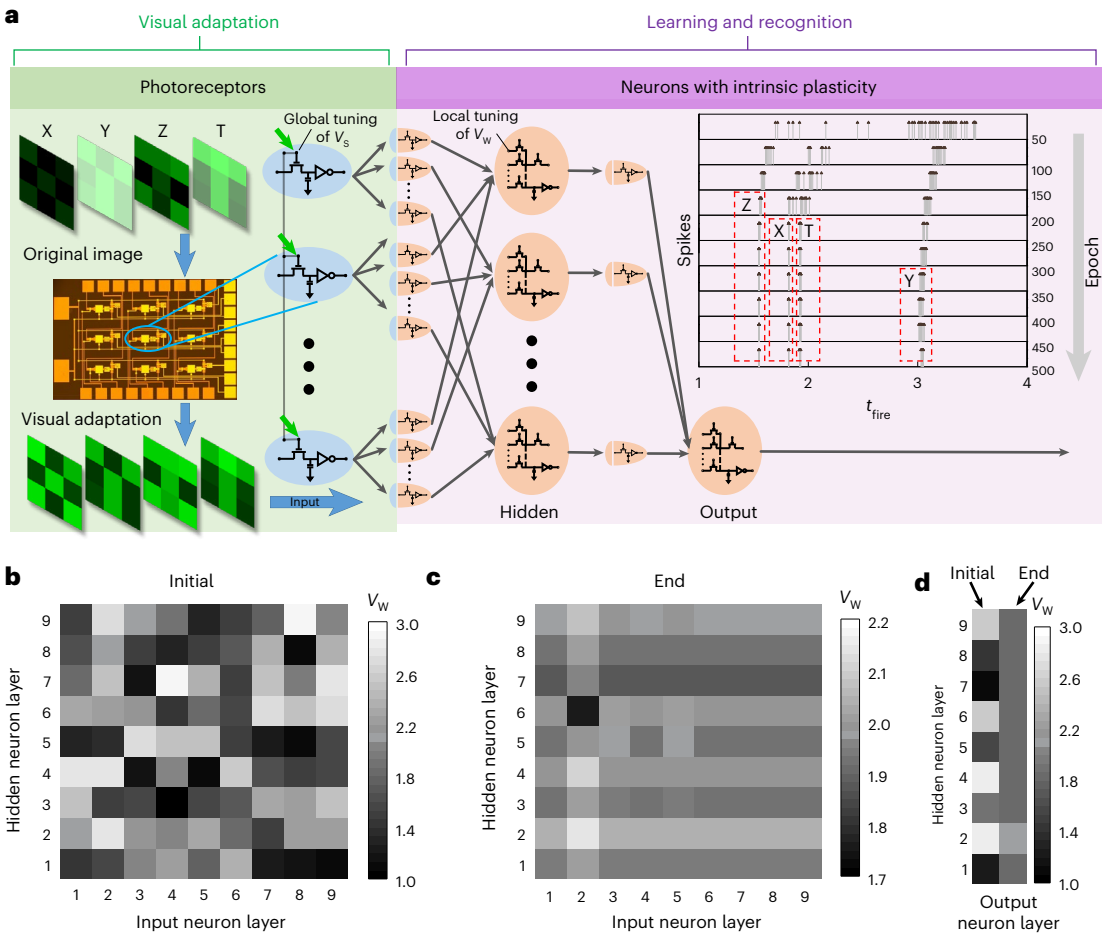


Fig. 5 | BioNN circuit for visual adaptation and feature recognition. a, Complete BioNN architecture of the in situ image processing and STDP learning simulation. **b**,**c**, The original randomly set V_w of the input-to-hidden layer (**b**) and the corresponding final states (**c**) after 500 epochs of learning. **d**, Initial and final states of V_w in the hidden-to-output layer.

we obtained the STDP learning rules of our neuron after simple mathematical transformations:

$$\Delta V_W = \begin{cases} \eta \times [\min(V_W) - V_W] \times e^{-|\Delta t|/\tau_+} & \text{if } \Delta t = t_{post} - t_{pre} > 0 \\ \eta\sigma \times [\max(V_W) - V_W] \times e^{-|\Delta t|/\tau_-} & \text{if } \Delta t = t_{post} - t_{pre} < 0 \end{cases}, \quad (4)$$

where the learning rate η is 0.01, the resting potential V_W is 2 V and the synapse time constant $\tau_+ = 5$ and $\tau_- = 5$. Supplementary Fig. 27 shows the relationship between the neuronal resting potential update ΔV_W and Δt modulated by the STDP learning rules. A 9 × 1 perceptron was first designed to test the effectiveness of the STDP learning rules on our BioNN circuit for unsupervised learning of the letter X, represented by a 3 × 3 pixel^2 array. Supplementary Fig. 28 shows the schematic of the 9 × 1 perceptron we simulated, along with the changes in neuronal membrane potential and corresponding t_{fire} during training. The complete learning process is shown in Supplementary Video 1. Supplementary Figs. 29 and 30 show the electrical signals of the neuron circuit in the first and last five epochs, respectively. The algorithm flowchart of our perceptron circuit simulation is illustrated in Supplementary Fig. 31, with all parameters obtained from experiments. All the modelling and simulation programs were written in MATLAB 8.4.

During processing, one letter was randomly selected from X, Y, Z and T with 10% random noise. In each epoch, the photoreceptor neurons adapted and encoded the letter to spike as the input of the BioNN. After performing 500 epochs of unsupervised learning based on STDP learning rules, Fig. 5b,c illustrates the initial random V_W and the corresponding final states after 500 epochs of learning, respectively. The grey level of the pixels indicates the resting potential of the neurons connecting the input and hidden neurons. Supplementary Video 2 records the evolution of V_W during the learning process. Figure 5d presents the initial and final states of V_W in the hidden-to-output layer. The evolution curves of V_W in both input-to-hidden and hidden-to-output layers are displayed in Supplementary Fig. 32. After 300 epochs, the network reached a well-learned state in which all V_W values converged to certain values. Figure 5a confirms this conclusion with the evolution of t_{fire}. Over the last 100 epochs, the t_{fire} values of the same letter have clustered together at discernible intervals from the other letters. This demonstrates the successful completion of the unsupervised learning task by transforming spatial information into temporal information. Supplementary Table 2 shows a review and summary of previous studies that have demonstrated intrinsic plasticity in different neuromorphic devices or circuits. Compared with similar previous works, our proposed neuron offers a broader versatility, higher encoding efficiency and better biological interpretability.

Conclusions

We have reported the development of an IF neuron module based on a DRAM inverter structure. We mimicked the membrane potential using the tunable voltage in the DRAM capacitor. Sensing and preprocessing

of optical stimuli was achieved through the tunable photoconductive effect of the photoreceptor (the MoS_2 transistor). The optical information was encoded via a TTFS coding mode. The adjustment of light sensitivity allowed our neuron module to adapt to different illumination backgrounds, achieving both photopic and scotopic adaptation. We also fabricated a 3 × 3 photoreceptor neuron array and demonstrated light coding and visual adaptation.

The modulation of neuronal intrinsic plasticity was emulated by tuning the neuronal resting potential V_W. We derived the STDP learning rules for high-level information processing and we used the experimental data to develop a photoelectric and electrical model of our neuron module. We used this model to construct a BioNN network simulation using MATLAB. The current–voltage behaviour simulations of the circuits were performed based on the neuronal intrinsic plasticity. Our simulated BioNN was able to recognize four letters within 300 epochs of training. Thus, neuronal intrinsic plasticity can be used to construct BioNNs, and could be an alternative way to realize a biologically inspired artificial visual systems.

Methods

Device fabrication

Device fabrication started with laser direct writing (MicroWriter ML3) to pattern the source/drain contact and bottom capacitor electrodes on a monolayer of MoS_2 film grown on a sapphire substrate. This was followed by electron-beam evaporation to deposit a 35-nm Au layer. The MoS_2 channel region was defined by using inductively coupled plasma etching. Next, a seed layer (4-nm SiO_2) was evaporated on the MoS_2 film using electron-beam evaporation, and then, a layer of 20-nm HfO_2 was deposited using atomic layer deposition to provide a high-κ dielectric. The dielectric layer acts as the gate dielectric in MoS_2 field-effect transistors and as the insulator layer for the capacitor. Inductively coupled plasma was used to form a via in the dielectric layer as an interconnect opening to connect the source in T_1 to the capacitor pad. Finally, the patterning and deposition processes were repeated to form the gate electrode of the MoS_2 field-effect transistors and the top electrode of the capacitor.

Electrical measurement

All the measurements were performed in the ambient environment at room temperature. For capacitor characterization, capacitance–voltage curves were measured using a Keysight E4990A impedance analyser. The MoS_2 transistors were characterized using a semiconductor parameter analyser (Agilent B1500A). For the DRAM and inverter measurements, the B1500A device was used to supply the voltage and detect the channel current. A waveform generator (Agilent 33600A) was used to supply waveforms to the test circuit, whereas an oscilloscope (DS 1054Z) was used to capture the output signal voltage.

BioNN simulation

Simulations were implemented in MATLAB on a 9 × 1 perceptron with parameters from experiments. Letters X, Y, Z and T (3 × 3 pixel^2) with 10% noise were presented over 500 epochs. Neuronal resting potentials V_W (1–3 V, initially randomized) were updated via STDP rules ($\eta = 0.01$, $\tau_+ = \tau_- = 5$). Convergence of V_W and clustering of t_{fire} confirmed successful unsupervised learning. Supplementary Information details the algorithm flow and temporal evolution.

Data availability

The source data file is available via figshare at https://figshare.com/s/a8fa70a5665cab68512d. Source data are provided with this paper.

Code availability

The codes used for simulation and data plotting are available from the corresponding authors upon reasonable request.

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Author contributions

Y.W., Y.C., P.Z. and W.B. conceived and supervised the project. S.G. introduced the concept of intrinsic plasticity. S.G., X.D., X.G., L.T. and J.M. fabricated the device. Y.W. and X.D. performed the BioNN simulation. Y.W., S.G., X.C., X.W. and Q.S. performed the electrical measurement. All authors discussed the results and contributed to the manuscript preparation.

Competing interests

The authors declare no competing interests.

Additional information

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